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Patent Public Search | Text View

United States Patent Application Publication

20250283299

Kind Code

A1

Publication Date

September 11, 2025

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WORKING MACHINE

Abstract

A working machine comprising: a machine body; a ground engaging propulsion structure supporting the machine body; a working arm pivotable relative to the machine body between raised and lowered positions; and an electrically powered prime mover arrangement configured to provide motive power to the ground engaging propulsion structure and/or the working arm. The machine body comprises: a working arm housing for receiving the working arm in its lowest position, the working arm housing comprising a sidewall; and an enclosure in which at least part of the prime mover arrangement is housed. Said at least part of the prime mover arrangement in the enclosure is adjacent the sidewall. The sidewall comprises an inlet aperture. The working machine is configured such that ambient air that passes through the inlet aperture, then passes over said at least part of the prime mover arrangement for cooling one or more electrical components thereof.

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Family ID: 1000008508631

Appl. No.: 19/073919

Filed: March 07, 2025

Foreign Application Priority Data

GB

2403332.6

Mar. 07, 2024

Publication Classification

Int. Cl.: E02F9/08 (20060101); B60K1/00 (20060101); B60L50/60 (20190101); E02F3/96 (20060101)

U.S. Cl.:

Background/Summary

FIELD

[0001] The present teachings relate to a working machine, and in particular to a working machine including an electrically powered prime mover arrangement.

BACKGROUND

[0002] Working machines are typically diesel-powered. The diesel engines of such working machines are typically cooled via a liquid cooling system in which heat is transferred from the engine to a radiator via a liquid coolant. The heat is then transferred to ambient air flowing along a cooling path over the radiator. Ambient air may flow along the cooling path as a result of operation of a fan and/or movement of the working machine. Although less common, the engines of some working machines may be exclusively air cooled, where heat is transferred directly from the engine to ambient air flowing along a cooling path over the engine. Typically, water in the environment (e.g., ground surface water or precipitation) is free to enter such cooling paths, since internal combustion engines and radiators are generally impervious to water.

[0003] There is a drive in the industry to move towards hybrid or electric powered vehicles. Like diesel-powered engines, the prime mover arrangements (e.g., including electric motors) of electrically powered working machines require cooling so as to prevent overheating. However, such electrical components are required to be shielded from water contact to prevent water-related damage thereto (e.g., short circuiting, corrosion).

[0004] To shield them from contact with water, electrical components of such prime mover arrangements are typically housed within an enclosure, such as a pod mounted to the vehicle's chassis, and typically cooled via air passing over the exterior surface of the enclosure. However, for larger vehicle's requiring larger prime mover arrangements, such cooling may be insufficient to prevent electrical components of the prime mover arrangement from overheating.

[0005] The present teachings seek to overcome or at least mitigate one or more problems associated with the prior art.

SUMMARY

[0006] The present teachings provide a working machine according to the appended claims.

[0007] An aspect of the teachings provides a working machine, e.g., a telehandler. The working machine may comprise a machine body. The working machine may comprise a ground engaging propulsion structure supporting the machine body. The working machine may comprise a working arm pivotable relative to the machine body between raised and lowered positions. The working machine may comprise an electrically powered prime mover arrangement configured to provide motive power to the ground engaging propulsion structure and/or the working arm. At least part of the prime mover arrangement may be housed within the machine body. A first external surface of the machine body may comprise an inlet aperture. The working machine may be configured such that ambient air that passes through the inlet aperture, then passes over said at least part of the prime mover arrangement for cooling one or more electrical components thereof. The first external surface may be configured so as to inhibit precipitation ingress to the cooling path via the inlet aperture.

[0008] Advantageously, such a configuration of the working machine helps to prevent overheating of the prime mover arrangement, and helps to inhibit precipitation from entering the enclosure and contacting the prime mover arrangement, which could lead to failure of the prime mover

arrangement (e.g., via short circuiting).

[0009] The machine body may comprise a working arm housing for receiving the working arm in its lowest position. The working arm housing may comprise a sidewall. The machine body may comprise an enclosure in which said at least part of the prime mover arrangement is housed. Said at least part of the prime mover arrangement in the enclosure may be adjacent the sidewall. The sidewall may comprise the inlet aperture.

[0010] Providing the inlet aperture in the sidewall of the working arm housing, as opposed to a front facing wall of the machine body for example, helps to inhibit precipitation from entering the enclosure and contacting the prime mover arrangement.

[0011] The inlet aperture may extend through an external surface of the sidewall. An angle between normals of the external surface and a ground plane of the working machine may be in the range of 75 to 105 degrees.

[0012] Advantageously, such an orientation of the external surface may help to inhibit precipitation ingress to the cooling path since precipitation is inhibited from falling directly through the inlet aperture.

[0013] The external surface may be substantially perpendicular to the ground plane.

[0014] The sidewall may be substantially parallel to a forward travelling direction of the working machine and/or a longitudinal axis of the working arm.

[0015] Advantageously, such a configuration helps to improve the packaging of the working arm housing.

[0016] The sidewall may be substantially perpendicular to a ground plane of the working machine.

[0017] Advantageously, such a configuration helps to improve the packaging of the working arm housing.

[0018] The enclosure may project laterally from the sidewall with respect to a forward travelling direction of the working machine.

[0019] Advantageously, such a configuration helps to improve packaging of the prime mover arrangement.

[0020] The inlet aperture may be adjacent to, or below, the working arm when the working arm is in its lowest position.

[0021] Advantageously, such a configuration helps to improve packaging of the working machine.

[0022] The working arm may comprise an upper wall and a lower wall connected by two opposed arm sidewalls. The inlet aperture may be adjacent to, or below, one of the arm sidewalls when the working arm is in its lowest position.

[0023] The inlet aperture may comprise a grill for inhibiting debris ingress to the enclosure.

[0024] At least part of the enclosure may be mounted to the sidewall.

[0025] Advantageously, such a configuration helps to improve packaging and assembly of the working machine.

[0026] Said at least part of the prime mover arrangement may be mounted to the sidewall.

[0027] Advantageously, such a configuration helps to improve packaging of the prime mover arrangement. Moreover, the first wall may act as heat sink, helping to further cool the prime mover arrangement.

[0028] The working arm may further comprise a duct within the enclosure configured to direct ambient air that passes through the inlet aperture towards said at least part of the prime mover arrangement.

[0029] Advantageously, the duct helps to maximize airflow over the drive arrangement to maximize cooling thereof.

[0030] The enclosure may comprise a first outlet aperture. Ambient air that passes through the inlet aperture, may pass over said at least part of the prime mover arrangement, and then through the first outlet aperture.

[0031] Advantageously, such a configuration helps to reduce the length of the cooling path.

[0032] The first outlet aperture may comprise a grill for inhibiting debris ingress to the enclosure.

[0033] The first outlet aperture may extend through an external surface of the enclosure. An angle between normals of the external surface and a ground plane of the working machine may be in the range of 90 to 180 degrees.

[0034] Advantageously, such a configuration helps to inhibit precipitation ingress to the enclosure since precipitation is inhibited from falling directly through the outlet aperture and into the enclosure.

[0035] The first outlet aperture may extend through an external surface of the enclosure. An angle between a normal of the external surface and a forward travelling direction of the working machine may be in the range of 90 to 180 degrees.

[0036] Advantageously, such a configuration helps to inhibit precipitation ingress to the enclosure since the ambient air is required to change direction to enter the enclosure via the outlet aperture.

[0037] The enclosure may comprise a wheel housing for a ground engaging wheel of the ground engaging propulsion structure. The wheel housing may comprise the first outlet aperture.

[0038] Advantageously, such a configuration help to improve packaging of the working
[0039] machine.

[0040] The first outlet aperture may face the ground engaging wheel. The working machine may comprise a barrier, such as a fender, interposed between the first outlet aperture and the ground engaging wheel for shielding the first outlet aperture from ground spray.

[0041] Advantageously, the barrier helps to prevent the ground engaging wheel spraying mud into the enclosure via the outlet aperture.

[0042] Ambient air that passes through the inlet aperture, over said at least part of the prime mover arrangement, and through the first outlet aperture, may travel along a substantially linear path.

[0043] Advantageously, such a linear path helps maximize airflow past the prime mover arrangement to maximize cooling thereof.

[0044] The working machine may further comprise a second outlet aperture in the machine body. The second outlet aperture may be spaced from the enclosure. Ambient air that passes through the inlet aperture, may then pass over at least part of the prime mover arrangement, and may then pass through the second outlet aperture.

[0045] Advantageously, such a configuration helps to maximize airflow over the prime mover arrangement for cooling thereof.

[0046] The sidewall may comprise an opening spaced from the inlet aperture. Ambient air that passes through the inlet aperture, may pass through said opening and may then pass through the second outlet aperture.

[0047] Advantageously, such a configuration helps to maximize airflow over the prime mover arrangement for cooling thereof.

[0048] Part of the prime mover arrangement (e.g., an electric motor) may extend through said opening.

[0049] Advantageously, such a configuration helps to improve packaging of the prime mover arrangement.

[0050] The working machine may further comprise at least one blower within the enclosure configured to force ambient air towards said at least part of the prime mover arrangement.

[0051] Advantageously, the blower may help to increase the flow of ambient air over the prime mover arrangement, increasing the cooling thereof.

[0052] The at least one blower may be adjacent the inlet aperture. The at least one blower may be selectively controllable so as to blow air out of the enclosure via the inlet aperture for blowing debris off a grill covering the inlet aperture.

[0053] Said at least part of the prime mover arrangement may comprise one or more inverters housed within an inverter housing. Ambient air that passes through the inlet aperture, may pass through the inverter housing for cooling the one or more inverters.

[0054] Said at least part of the prime mover arrangement may comprise at least one heatsink connected to at least one electrical component, such as an inverter. Ambient air that passes through the inlet aperture, may pass over said at least one heatsink.

[0055] Advantageously, the at least one heat sink may help to more efficiently cool the prime mover arrangement.

[0056] The prime mover arrangement may comprise at least two heatsinks, each connected to at least one electrical component of the prime mover arrangement. Said at least two heatsinks may be arranged in parallel with respect to a flow path of ambient air that passes through the inlet aperture, and over said at least part of the prime mover arrangement.

[0057] Advantageously, such a configuration of the at least two heat sinks may help to more efficiently cool the prime mover arrangement.

[0058] Ambient air that passes through the inlet aperture may cool one or more of the following electrical components of the prime mover arrangement: one or more inverters; one or more electric motors; one or more electric pumps; and one or more power distribution units.

[0059] The enclosure may comprise an access port for accessing the prime mover arrangement. The enclosure may comprise a hatch movable relative to the access port between open and closed positions for respectively opening and closing the access port. The hatch may inhibit precipitation ingress to the enclosure via the access port when in the closed position.

[0060] Advantageously, provision of the access port and hatch enables access to the prime mover arrangement for maintenance and inspection whilst inhibiting precipitation contacting the prime mover arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0061] Embodiments are now disclosed by way of example only with reference to the drawings, in which:

[0062] FIGS. **1a** shows an isometric view of a working machine according to an embodiment;

[0063] FIG. **1b** show a side view of the working machine of FIG. **1**;

[0064] FIG. **2** shows a front isometric view of a prime mover arrangement and an enclosure of the working machine of FIG. **1a**;

[0065] FIG. **3** shows a front isometric view of the prime mover arrangement of FIG. **2** with the enclosure removed;

[0066] FIG. **4** shows a further front isometric view of the prime mover arrangement of FIG. **2** with the enclosure removed;

[0067] FIG. **5a** shows a front isometric view of the prime mover arrangement of FIG. **3** with a duct cover removed;

[0068] FIG. **5b** shows a front isometric detail view of the prime mover arrangement of FIG. **5a** with an inverter cover removed;

[0069] FIG. **6** is a topside plan view of the prime mover arrangement of FIG. **5a**;

[0070] FIG. **7** is a rear isometric view of the prime mover arrangement and enclosure of FIG. **2**;

[0071] FIG. **8** is a front isometric view of a chassis, motor and cab of the working machine of FIG. **1a**;

[0072] FIG. **9** is an underside plan view of the chassis of FIG. **8**; and

[0073] FIG. **10** shows a front isometric view of a prime mover arrangement and an enclosure of a working machine according to an embodiment.

DETAILED DESCRIPTION OF EMBODIMENT(S)

[0074] In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of various embodiments and the teachings. However, those

skilled in the art will understand that: the present teachings may be practiced without these specific details or with known equivalents of these specific details; that the present teachings are not limited to the described embodiments; and, that the present teachings may be practiced in a variety of alternative embodiments. It will also be appreciated that well known methods, procedures, components, and systems may not have been described in detail.

[0075] References to vertical and horizontal in the present disclosure should be understood to be in relation to the machine when stood on horizontal ground in a non-working condition.

[0076] With reference to FIGS. **1a** to **9** there is shown a working machine **10**. In the illustrated embodiment, the working machine **10** is a telescopic handler (i.e., telehandler). In alternative embodiments (not shown), the working machine **10** may be any suitable working machine, such as a backhoe loader, a forklift truck, a wheeled loading shovel, a dumper, an excavator or a tractor, for example. Such working machines **10** are suitable for use in off-highway industries such as agriculture and construction. In these industries they are generally configured to perform tasks such as excavation, load handling, harvesting or planting crops.

[0077] The working machine **10** includes a machine body **12**, an operator cab **13**, a working implement assembly **14**, and a ground engaging propulsion structure **18**.

[0078] The working implement assembly **14** is configured to handle material. In the illustrated embodiment, the working implement assembly **14** includes a telescopic working arm **14A** pivotably mounted at a first end to the body **12** for pivoting movement relative to the body **12** about a first generally horizontal axis A (represented as a dot-dash line in FIG. **1a**) between lowered and raised positions. The working arm **14A** is configured to carry a load via a handling implement (not shown) at a distal second end opposite to the proximal first end. In the illustrated embodiment, the second end includes a tool carrier **20** (or a carriage) which is configured to attach to or comprises a working implement/work tool to the machine **10**. The tool carrier **20** is configured to pivot relative to the working arm **14A** about a second generally horizontal axis B (which may be referred to as a tilt axis B) so that a load may be kept in a constant horizontal or other chosen orientation as the working arm **14A** pivots up and down, as well known in the art. In alternative embodiments, the working implement assembly **14** may include any suitable working implement (e.g., any suitable (e.g., non-telescopic) working arm).

[0079] The ground engaging propulsion structure **18** supports the body **12**, and is configured to provide a tractive effort to propel the machine **10** over ground. The propulsion structure **18** of the example shown in the figures includes a front axle AF and a rear axle AR. A pair of front wheels **18F** are coupled to the front axle AF, and a pair of rear wheels **18R** are coupled to the rear axle AR. The wheels **18F**, **18R** may be configured to provide two wheel or four-wheel steering as also well known in the art. Either or both of the front and rear wheels **18F**, **18R** may be configured to turn relative to the body **12** under the influence of a steering device provided within the operator cab **13** to effect steering. In the illustrated embodiment, the steering device includes a steering wheel **26** in the operator cab **13**, but may additionally or alternatively include any suitable device such as a joystick for example. In alternative embodiments, the ground engaging propulsion structure **18** may include a pair of endless tracks.

[0080] With reference to FIGS. **3** and **4**, the working machine **10** includes an electrically powered prime mover arrangement **30** configured to provide motive power to the propulsion structure **18** and the working implement assembly **14**. In the illustrated embodiment, the prime mover arrangement **30** includes an electric motor **32** for providing motive power to the propulsion structure **18**, and an electric pump **33** for providing motive power to the working implement assembly **14**. The electric pump **33** may include an electric motor (e.g., for rotating an impeller of the pump) separate to the electric motor **32**.

[0081] In the illustrated embodiment, both of the axles AF, AR are coupled to the electric motor **32** via a transmission so as to drive movement of the ground engaging propulsion structure **18** (i.e., the axles AF, AR). The machine **10** of this embodiment is thus four wheel drive. The electric motor **32**

causes movement of the working machine **10** over a ground surface. In alternative embodiments, only one of the axles AF, AR may be coupled to the electric motor **32** (i.e., the machine **10** may be two wheel drive).

[0082] In the illustrated embodiment, the prime mover arrangement **30** includes a single electric motor **32**, but in alternative embodiments may include more than one electric motor (e.g., each wheel of the propulsion structure **18** may be driven by one or more dedicated electric motors). In some embodiments, the prime mover arrangement **30** may include a hybrid including both an internal combustion engine and an electric motor for providing motive power to the propulsion structure **18**.

[0083] In the illustrated embodiment, the transmission includes a “dropbox” or gear train by which power is transferred from the electric motor **32** to the propulsion structure **18**.

[0084] In the illustrated embodiment, the electric pump **33** provides motive power to the working implement assembly **14** to raise and lower the working arm **14A** and pivot the tool carrier **20** via a hydraulic actuation system. The electric pump **33** is configured to supply pressurized hydraulic fluid to the hydraulic actuation system for moving the working implement assembly **14**. In alternative embodiments (not shown), the prime mover arrangement **30** may additionally or alternatively include a different electrically powered prime mover for moving the working implement assembly **14**, such as one or more electric motors. In some embodiments, the propulsion structure **18** and the working implement assembly **14** may be driven via a single common electric pump or motor.

[0085] In alternative embodiments (not shown), the prime mover arrangement **30** may provide motive power to only one of the propulsion structure **18** and working implement assembly **14**.

[0086] The working machine **10** includes an electric energy storage arrangement (e.g., including one or more batteries) (not shown) for powering the prime mover arrangement **30**. In the illustrated embodiment, the electric energy storage arrangement is housed within a portion **15** of the body **12** aft of the cab **13** (see FIG. **1a**). In alternative embodiments, the electric energy storage arrangement may be located in any suitable portion of the body **12** (e.g., within the enclosure **36** outlined below).

[0087] The body **12** includes a chassis **34** which provides the main structural support for the working machine **10**. The chassis **34** may include structural support for the propulsion structure **18**, the prime mover arrangement **30**, the operator cab **13**, the working implement assembly **14** and associated actuators, and any ancillary equipment, components, systems or body work as may be required for the machine **10** to function. The operator cab **13** is mounted to the chassis **34**.

[0088] In the illustrated embodiment, the chassis **34** includes a working arm housing **37** having a first or outer sidewall **34_1**, a second or inner sidewall **34_2**, and a base wall **34_3**. The working arm **14A** is received in the working arm housing **37** when the working arm **14A** is in its lowest position. At least part of the prime mover arrangement **30** is adjacent the outer sidewall **34_1**.

[0089] With reference to FIG. **2**, the body **12** includes an enclosure **36** mounted to the chassis **34**. At least part of the prime mover arrangement **30** is housed in the enclosure **36**. The enclosure **36** helps to protect the prime mover arrangement **30** from external damage, and shields the prime mover arrangement **30** from precipitation. In the illustrated embodiment, the enclosure **36** is configured to inhibit precipitation ingress within the enclosure **36**. Advantageously, this helps to inhibit water contacting electrical components of the prime mover arrangement **30**, which could lead to damage occurring to the prime mover arrangement **30** (e.g., short circuiting and/or corrosion).

[0090] In the illustrated embodiment, the enclosure **36** is in the form of a pod projecting from a lateral side of the chassis **34**. The cab **13** is offset to one lateral side of a forward-aft centerline of the body **12**. The enclosure **36** projects from the opposite (i.e., non-cab) lateral side of the chassis **34**. The enclosure **36** is interposed between the front and rear wheels **18F**, **18R** on said lateral side of the chassis **34**. In alternative embodiments, the enclosure **36** may have any suitable positioning

on the working machine **10**.

[0091] In the illustrated embodiment, the enclosure **36** includes an outer shell **38** mounted to the outer sidewall **34_1**, such that the outer shell **38** and the outer sidewall **34_1** define the enclosure **36**. The enclosure **36** projects laterally from the outer sidewall **34_1** (i.e., in a direction generally transverse to a forward travelling direction F of the working machine **10**). The outer sidewall **34_1** is interposed between the enclosure **36** and the working arm **14A** when in its lowest position as shown in FIGS. **1a** and **1b**.

[0092] The enclosure **36** includes an access port **36_1**, and a hatch **36_2** movable relative to the access port **36_1** between open and closed positions for respectively opening and closing the access port **36_1**. The prime mover arrangement **30** is accessible via the access port **36_1** (e.g., for inspection and maintenance) when the hatch **36_2** is in the open position. The hatch **36_2** inhibits precipitation ingress to the enclosure **36** via the access port **36_1** when in the closed position.

[0093] In the illustrated embodiment, an upwardly facing outer surface of the enclosure **36** includes the access port **36_1**. The hatch **36_2** is positioned above the access port **36_1**. The hatch **36_2** is pivotable relative to the access port **36_1** between open and closed positions. In the open position, the hatch **36_2** extends over at least part of the prime mover arrangement **30**, so as to shield it from precipitation as shown in FIGS. **2** and **7**.

[0094] The enclosure **36** may include a seal member (e.g., formed from a flexible polymeric material) configured to form a seal between the access portion **36_1** and the hatch **36_2** when the hatch **36_2** is in closed position. For example, the seal member may be mounted around the upwardly facing outer surface of the enclosure **36**.

[0095] The prime mover arrangement **30** may include a number of electrical components for supplying power to and/or controlling the motor **32** and electric pump **33**. In the illustrated embodiment, the prime mover arrangement **30** includes the electric energy storage arrangement, and an inverter arrangement **30_1** including one or more power inverters. The inverter arrangement **30_1** converts direct current supplied from the electric energy storage arrangement to alternating current for supply to the motor **32** and the pump **33**.

[0096] In the illustrated embodiment, the electric motor **32**, the electric pump **33**, and the inverter arrangement **30_1** are at least partially housed within the enclosure **36**.

[0097] It will be appreciated that, during operation of the working machine **10**, waste heat generated by the electrical components of the prime mover arrangement **30** will cause their temperature to increase. To prevent electrical components of the prime mover arrangement **30** from overheating, the working machine **10** includes a cooling system **40** for cooling said electrical components.

[0098] With reference to FIGS. **5a** to **7**, the cooling system **40** includes a cooling path **42** (represented by dashed double line arrows in FIGS. **5a**, **6** and **7**) for transporting ambient air over at least part of the prime mover arrangement **30** for cooling one or more electrical components thereof. The cooling path **42** passes from an inlet aperture **44** in a first external surface of the body **12**, over part of the prime mover arrangement **30** (hereinafter “the cooled part of the prime mover arrangement”), and then through an outlet aperture **46** in a second external surface of the body **12**. Advantageously, the cooling path **42** helps to prevent overheating of the prime mover arrangement **30**.

[0099] In the illustrated embodiment, the inlet aperture **44** has a substantially triangular profile, but may have any suitable shape in alternative embodiments.

[0100] In the illustrated embodiment, the cooling path **42** passes through the enclosure **36**. To prevent water contacting electrical components of the prime mover arrangement **30**, the cooling path **42** is configured to inhibit precipitation ingress to the cooling path **42**, and thus to the enclosure **36**.

[0101] In the illustrated embodiment, the outer sidewall **34_1** includes the inlet aperture **44**. The inlet aperture **44** passes through the outer sidewall **34_1** from an external surface **50** thereof to an

internal surface **52** thereof. Although not visible in the figures, it will be appreciated that the external surface **50** faces the inner sidewall **34_2**. By “external surface” it is intended to mean a surface on the exterior of the working machine **10**, which is exposed to the external environment. By “internal surface” it is intended to mean a surface on the interior of the working machine **10**, which is shielded from the external environment. The internal surface **52** is within the enclosure **36**, and thus on the interior of the working machine **10**. In the following, the wording “the external surface **50**” shall refer to the external surface of the working machine **10** immediately surrounding the inlet aperture **44**.

[0102] In the illustrated embodiment, the outer sidewall **34_1** is formed as a (e.g., metal) plate, and is thus substantially planar. In alternative embodiments (not shown), any suitable planar or non-planar wall may include the inlet aperture **44**.

[0103] In the illustrated embodiment, the outer sidewall **34_1** is a longitudinal wall (i.e., it is substantially parallel to the forward travelling direction F and/or a longitudinal axis of the working arm **14A**).

[0104] In the illustrated embodiment, the outer sidewall **34_1** is substantially perpendicular to a ground plane of the working machine **10**.

[0105] In the illustrated the inlet aperture **44** is adjacent the working arm **14A** in its lowest position. A gap between the working arm **14A** and the inlet aperture **44** in its lowest position enables ambient air to flow from outside of the working machine **10** into the cooling path **42** via the inlet aperture **44**. In alternative embodiments (not shown), the inlet aperture **44** may be below the working arm **14A** when the working arm **14A** is in its lowest position.

[0106] In the illustrated embodiment, the working arm **14A** includes an upper wall **14U** (see FIG. **1a**) and an opposite lower wall (not visible in the figures) connected by two opposed arm sidewalls **14S** (only one of which is shown in FIG. **1a**). The inlet aperture **44** is adjacent to, or below, one of the arm sidewalls **14S** when the working arm **14A** is in its lowest position.

[0107] In the illustrated embodiment, the electric motor **32**, and the inverter arrangement **30_1** are mounted to the outer sidewall **34_1**. The outer sidewall **34_1** may thus act as a heat sink for these components, helping to further cool the prime mover arrangement **30**.

[0108] The external surface **50** of the outer sidewall **34_1** is configured so as to inhibit precipitation ingress to the cooling path **42** via the inlet aperture **44**. Advantageously, such a configuration of the external surface **50** helps to enable air to flow along the cooling path **42** so as to prevent overheating of the prime mover arrangement **30**, whilst inhibiting water from contacting electrical components of the prime mover arrangement **30**.

[0109] In the illustrated embodiment, the external surface **50** including the inlet aperture **44** is substantially perpendicular to a horizontal ground plane P of the working machine **10**, i.e., a normal (i.e., normal line or vector) of the external surface **50** is substantially perpendicular to a normal of the ground plane P. Precipitation is thus inhibited from falling directly through the inlet aperture **44** and into the cooling path **42**. In alternative embodiments, an angle between normals of the external surface **50** including the inlet aperture **44** and the ground plane P may be in the range of 75 to 105 degrees.

[0110] In alternative embodiments (not shown), the external surface **50** may include a canopy arranged above the inlet aperture **44** so as to inhibit precipitation ingress to the cooling path **42** via the inlet aperture **44**.

[0111] The external surface **50** is configured so as to divert ambient air prior to entering the cooling path **42** via the inlet aperture **44**. Advantageously, such diverting of the ambient air may help to inhibit water entrained with the ambient air from entering the cooling path **42** via the inlet aperture **44**.

[0112] In the illustrated embodiment, the external surface **50** is substantially parallel to a forward travelling direction F of the working machine **10**, i.e., the normal of the external surface **50** is substantially perpendicular to the forward travelling direction F. When the working machine **10** is

travelling over ground in the forward travelling direction F, air travelling over the working machine **10** in a direction opposite to the forward travelling direction F (in a reference frame fixed to the working machine **10**) is thus necessarily diverted prior to entering the cooling path **42** via the inlet aperture **44**. In alternative embodiments, an angle between the normal of the external surface **50** including the inlet aperture **44** and the forward travelling direction F may be in the range of 90 to 180 degrees (i.e., such that the external surface **50** may face away from the forward travelling direction F).

[0113] In alternative embodiments (not shown), the external surface **50** may include a baffle arrangement configured so as to divert ambient air prior to entering the cooling path **42** via the inlet aperture **44**.

[0114] With reference to FIGS. 2 to 4, the cooling system **40** includes a duct **54** within the enclosure **36** configured to direct ambient air entering the cooling path **42** via the inlet aperture **44**, towards the cooled part of the prime mover arrangement **30**. The cooling path **44** passes through the duct **54**. In the illustrated embodiment, the duct **54** is formed between the outer sidewall **34_1** and a duct cover **55** mounted to the outer sidewall **34_1** (e.g., via one or more fasteners, bonding and/or welding). The ambient air is directed towards the cooled part of the prime mover arrangement **30** by a smooth and continuous surface of the cover **55** so as to inhibit stagnation points from forming in the cooling path **42**.

[0115] A lower portion of the cover **55** may include a drain for draining water that has entered the duct **54** via the inlet aperture **44**.

[0116] The duct **54** separates a portion of the cooling path **42** upstream of the cooled part of the prime mover arrangement **30** from the remainder of the enclosure **36**. This helps to increase the flow of ambient air passing over the cooled part of the prime mover arrangement **30**. In the illustrated embodiment, said upstream portion of the cooling path **42** extends from the inlet aperture **44** to the cooled part of the prime mover arrangement **30**.

[0117] Turning again to FIGS. 5a and 5b, the cooling system **40** includes at least one blower configured to force ambient air along the cooling path **42** from the inlet aperture **44** towards the cooled part of the prime mover arrangement **30**. The at least one blower helps to increase the flow of ambient air over the cooled part of the prime mover arrangement **30**. In the illustrated embodiment, the at least one blower includes two electric fans **56**. In alternative embodiments (not shown), the cooling system **40** may include only one, more than two, or no fans. In alternative embodiments (not shown), the cooling system **40** may include any suitable blower for forcing ambient air along the cooling path **42** from the inlet aperture **44** to the cooled part of the prime mover arrangement **30**, other than a fan, such as a (e.g., electric) pump.

[0118] In the illustrated embodiment, the inverter arrangement **30_1** includes a first inverter **60_1** for supplying alternating current to the motor **32**, and a second inverter **60_2** for supplying alternating current to the pump **33**. In alternative embodiments (not shown), the inverter arrangement **30_1** may include only one or more than two inverters.

[0119] The inverter arrangement **30_1** includes a first heatsink **58_1** connected to the first inverter **60_1**, and a second heatsink **58_2** connected to the second inverter **60_2**. A major face of each inverter **60_1**, **60_2** is in contact with the respective heatsink **58_1**, **58_2** so as to increase heat transfer from the inverters **60_1**, **60_2** to the heatsinks **58_1**, **58_2**. The cooling path **42** passes over the heatsinks **58_1**, **58_2** so as to transfer heat from the heatsinks **58_1**, **58_2** to the ambient air flowing along the cooling path **42**. Such an arrangement helps to more efficiently cool the inverters **60_1**, **60_2**.

[0120] In the illustrated embodiment, each heatsink **58_1**, **58_2** is formed from a (e.g., metal) material having a high thermal conductivity, and includes a plurality of fins **62** for increasing the contact surface area with ambient air flowing along the cooling path **42**.

[0121] In alternative embodiments (not shown), the prime mover arrangement **30** may include only one, more than two, or no heatsinks. In some embodiments, one or more heatsinks may be

connected to any suitable electrical component(s) of the prime mover arrangement **30**.

[0122] In the illustrated embodiment, the heatsinks **58_1**, **58_2** are arranged in parallel with respect to the cooling path **42** (i.e., the ambient air flowing along the cooling path **42** passes over the heatsinks **58_1**, **58_2** in parallel rather than sequentially). This arrangement helps to cool the heatsinks **58_1**, **58_2** more efficiently. In alternative embodiments (not shown), the heatsinks **58_1**, **58_2** may be arranged at least partially in series.

[0123] In the illustrated embodiment, each fan **56** is upstream of one of the heatsinks **58_1**, **58_2**, which helps to increase the flow of ambient air over the heatsinks **58_1**, **58_2**, so as to increase the cooling thereof. In alternative embodiments, the fans **56** may have any suitable arrangement. For example, at least one of the fans **56** may be downstream of one or both of the heatsinks **58_1**, **58_2**.

[0124] In the illustrated embodiment, the inverter arrangement **30_1** is housed within an inverter housing **53**. The cooling path **42** passes through the inverter housing **53** for cooling the inverters **60_1**, **60_2** via cooling of the heatsinks **58_1**, **58_2**.

[0125] With reference to FIGS. **6** and **7**, downstream of the heatsinks **58_1**, **58_2**, the cooling path **42** passes over a voltage converter arrangement **64** and a power distribution unit **67**, and then passes through the outlet aperture **46**. The cooling system **40** may thus also cool other electrical components of the working machine **10** such as the voltage converter arrangement **64**. The voltage converter arrangement **64** and power distribution unit **67** are arranged in parallel with respect to the cooling path **42**, but may be arranged at least partially in series in alternative embodiments.

[0126] The voltage converter arrangement **64** converts direct current from the electric energy storage arrangement to one or more different voltages required by different electrical components of the working machine **10**. In the illustrated embodiment, the voltage converter arrangement **64** charges a **12** volt battery **35**, which powers one or more **12** volt electrical systems (e.g., lighting) of the working machine **10**. The **12** volt battery **35** is housed within the enclosure **36**.

[0127] In the illustrated embodiment, the cooling path **42** is substantially linear (i.e., extends along a substantially straight line), which helps to maximize airflow over the cooled part of the prime mover arrangement **30**. In alternative embodiments (not shown), the cooling path **42** may be non-linear (e.g., may include one or more bends).

[0128] In the illustrated embodiment, the enclosure **36** includes the outlet aperture **46**. In particular, a rear portion of the enclosure **36** forms a wheel housing **36_2** for one of the rear wheels **18R**, and the wheel housing **36_1** includes the outlet aperture **46**. The outlet aperture **46** passes through a wall **65** of the enclosure **36** from an internal surface thereof (not shown) to an external surface **66** thereof. In the following, the wording “the external surface **66**” shall refer to the external surface of the working machine **10** immediately surrounding the outlet aperture **46**.

[0129] In the illustrated embodiment, the outlet aperture **46** faces the rear wheel **18_R**. As shown in FIGS. **1a** and **1b**, to shield the outlet aperture **46** from ground spray churned up by the rear wheel **18_R**, the working machine **10** includes a barrier **68**, such as a fender, interposed between the outlet aperture **46** and the rear wheel **18_R**.

[0130] The external surface **66** including the outlet aperture **46** is configured so as to inhibit precipitation ingress to the cooling path **42** via the outlet aperture **46**. In the illustrated embodiment, an angle between normals of the external surface **66** and the ground plane **P** is in the range of 90 to 180 degrees, such that the external surface **66** faces towards the ground plane **P** of the working machine **10**. Precipitation is thus unable to fall directly through the outlet aperture **46** and into the cooling path **42**. In alternative embodiments (not shown), the external surface **66** may be substantially perpendicular to the ground plane **P**, like the external surface **50** which includes the inlet aperture **44**.

[0131] In the illustrated embodiment, the external surface **66** includes a canopy **69** arranged above the outlet aperture **46** so as to inhibit precipitation ingress to the cooling path **42** via the outlet aperture **46**.

[0132] The external surface **66** including the outlet aperture **46** is configured so as to inhibit

ambient air which is flowing over the working machine **10** from entering the cooling path **42** via the outlet aperture **46**, when the working machine **10** is travelling in the forward travelling direction **F**. In the illustrated embodiment, an angle between the normal of the external surface **66** and the forward travelling direction **F** is in the range of 90 to 180 degrees, such that the external surface **66** faces away from the forward travelling direction **F**. Air travelling over the working machine **10** in a direction opposite to the forward travelling direction **F** (in a reference frame fixed to the working machine **10**) is thus inhibited from entering the cooling path **42** via the outlet aperture **46**. In alternative embodiments (not shown), the external surface **66** including the outlet aperture **46** may be substantially parallel to the forward travelling direction **F**, like the external surface **50** including the inlet aperture **44**.

[0133] In alternative embodiments (not shown), the external surface **66** may include a baffle arrangement configured to inhibit ambient air which is flowing over the working machine **10** from entering the cooling path **42** via the outlet aperture **46**.

[0134] The inlet aperture **44** is located forward of the outlet aperture **46** with respect to the forward travelling direction **F** of the working machine **10**. Such an arrangement of the inlet and outlet apertures **44**, **46** inhibits warmed air exiting the outlet aperture **46** from entering the inlet aperture **44**, and thus helps to more efficiently cool the prime mover arrangement **30**. In the illustrated embodiment, the inlet aperture **44** is at or towards a forward end **36_F** of the enclosure **36**, and the outlet aperture **46** is at or towards a rear end **36_R** of the enclosure **36**.

[0135] With reference to FIGS. **8** and **9**, the cooling system **40** of the illustrated embodiment includes a further cooling path **70** (represented by a dashed line in FIG. **8**) for transporting ambient air over at least a further part of the prime mover arrangement **30** (relative to the cooling path **40**), for cooling one or more electrical components thereof. The further cooling path **70** passes from the inlet aperture **44**, over the further cooled part of the prime mover arrangement **30**, and through a second outlet aperture **72** in an external surface **74** of the body **12**. The second outlet aperture **72** is spaced from the enclosure **36**. Provision of the further cooling path **70** helps to maximize airflow over the prime mover arrangement **30** for cooling thereof, as well as help cool a greater proportion of the prime mover arrangement **30**.

[0136] In the illustrated embodiment, the further cooling path **70** passes over the inverter arrangement **30_1**, the pump **33** and the motor **32**, but may pass over any electrical component(s) of the prime mover arrangement **30** in alternative embodiments.

[0137] In the illustrated embodiment, the outer sidewall **34_1** includes an opening **76** spaced from the inlet aperture **44**. The further cooling path **70** passes from the enclosure **36**, through the opening **76** and then through the second outlet aperture **72**.

[0138] In the illustrated embodiment, the motor **32** extends through the opening **76**, which helps to improve packaging of the prime mover arrangement **30**. In alternative embodiments (not shown), other components of the prime mover arrangement **30** (e.g., the pump **33**) may additionally or alternatively extend through the opening **76**.

[0139] In the illustrated embodiment, an underside of the chassis **34** includes the second outlet aperture **72**. In alternative embodiments (not shown), any suitable portion of the body **12** may include the second outlet aperture **72**.

[0140] In alternative embodiments (not shown), the working machine **10** may include only one of the cooling path **42** and the cooling path **70**.

[0141] In some embodiments, the working machine **10** may be configured such that ambient air travelling along the cooling path **40** and/or the further cooling path **70** cools one or more of the following electrical components of the prime mover arrangement **30**: one or more inverters **60_1**, **60_2**; one or more electric motors **32**; and one or more electric pumps **33**.

[0142] The inlet aperture **44** and/or outlet aperture **46** may be at least partially covered by a grill to inhibiting the ingress of debris, dust and dirt to the enclosure **36**.

[0143] To inhibit dust and debris from blocking such a grill, the working machine **10** may be

configured to selectively control the fans **56** so as to blow air out of the enclosure **36** via the inlet aperture **44** for blowing dust or debris off the grill. In such embodiments, at least one of the fans **56** may be adjacent the inlet aperture **44**.

[0144] FIG. **10** shows part of a working machine **10'** according to a further embodiment. Features in common with the embodiment shown in FIGS. **1** to **9** share common reference numerals, and a discussion of these features will not be repeated for brevity. Unless stated otherwise, the working machine **10'** shown in FIG. **10** may share any of the features described in relation to the working machine **10** shown in FIGS. **1** to **9**.

[0145] The working machine **10'** includes an electrically powered prime mover arrangement **30'** having a different arrangement to the powered prime mover arrangement **30** of the previous embodiment. The working machine **10'** is larger than the working machine **10** of the previous embodiment. As such, components of the prime mover arrangement **30'** are larger relative to the previous embodiment, and are arranged differently so as to improve their packaging.

[0146] The prime mover arrangement **30'** includes an inverter arrangement **30_1'** including one or more heatsinks (not shown), each connected to an inverter (not shown) of the inverter arrangement **30_1'**. The inverter arrangement **30_1'** includes an inverter housing **53'** housing the one or more heatsinks. One or more inverters of the inverter arrangement **30_1'** may also be housed within the inverter housing **53'**.

[0147] The working machine **10'** has a cooling system **40'** including a cooling path **42'** (represented by dashed double line arrows in FIG. **10**) for transporting ambient air over at least part of the prime mover arrangement **30'** for cooling thereof. Initially, the cooling path **42'** passes from the inlet aperture **44** (shown in phantom in FIG. **10**) in a substantially lateral direction (with respect to the forward traveling direction F) into the inverter housing **53'** via one or more inverter housing inlets (not shown) adjacent the inlet aperture **44**, then over the one or more heatsinks housed therein, and then out of the inverter housing **53'** via one or more inverter housing outlets **92**. As such, the inverter housing **53'** acts as a duct within the enclosure **36**, directing the cooling path **42'** in a lateral direction from the inlet aperture **44**. The outer shell of the enclosure **36** then forces the cooling path **42'** exiting the one or more inverter housing outlets **92** to change direction towards the outlet aperture **46**, from which the cooling path **42'** exits the enclosure **36**. Such an arrangement of the cooling path **42'** enables a greater volume of air to flow over the one or more inverter heatsinks, helping to improve cooling thereof.

[0148] The working machine **10'** includes a heat exchanger **94** within the enclosure **36** and adjacent the outlet aperture **46**. The cooling path **42'** passes over the heat exchanger **94** before exiting the enclosure **36** via the outlet aperture **46**. The heat exchanger **94** is configured to cool hydraulic fluid of a hydraulic system of the working machine **10'** (e.g., for operating the working arm **14A**). Hydraulic fluid is pumped through the heat exchanger **94**, and cooled via the cooling path **42'**.

[0149] The heat exchanger **94** includes a fan (not shown) configured to force air along the cooling path **42'** and through the outlet aperture **46**. In alternative embodiments (not shown), the working machine **10'** may include one or more fans (or any suitable blower) adjacent the outlet aperture **46**, even when the heat exchanger **94** is not provided or located elsewhere.

[0150] Unless stated otherwise, the working machine **10** shown in FIGS. **1** to **9** may share any of the features described in relation to the working machine **10'** shown in FIG. **10**.

[0151] The one or more embodiments are described above by way of example only and it will be appreciated that the variations are possible without departing from the scope of protection afforded by the appended claims.

Claims

1. A working machine comprising: a machine body; a ground engaging propulsion structure supporting the machine body; a working arm pivotable relative to the machine body between raised

and lowered positions; and an electrically powered prime mover arrangement configured to provide motive power to the ground engaging propulsion structure and/or the working arm, wherein the machine body comprises: a working arm housing for receiving the working arm in its lowest position, the working arm housing comprising a sidewall; and an enclosure in which at least part of the prime mover arrangement is housed, wherein said at least part of the prime mover arrangement in the enclosure is adjacent the sidewall, wherein the sidewall comprises an inlet aperture, and wherein the working machine is configured such that ambient air that passes through the inlet aperture, then passes over said at least part of the prime mover arrangement for cooling one or more electrical components thereof.

2. The working machine of claim 1, wherein the inlet aperture extends through an external surface of the sidewall, wherein an angle between normals of the external surface and a ground plane of the working machine is in the range of 75 to 105 degrees; optionally, wherein the external surface is substantially perpendicular to the ground plane.

3. The working machine of claim 1, wherein the sidewall is substantially parallel to a forward travelling direction of the working machine and/or a longitudinal axis of the working arm.

4. The working machine of claim 1, wherein the sidewall is substantially perpendicular to a ground plane of the working machine.

5. The working machine of claim 1, wherein the enclosure projects laterally from the sidewall with respect to a forward travelling direction of the working machine.

6. The working machine of claim 1, wherein the inlet aperture is adjacent to, or below, the working arm when the working arm is in its lowest position; optionally, wherein the working arm comprises an upper wall and a lower wall connected by two opposed arm sidewalls, and wherein the inlet aperture is adjacent to, or below, one of the arm sidewalls when the working arm is in its lowest position.

7. The working machine of claim 1, wherein the inlet aperture comprises a grill for inhibiting debris ingress to the enclosure.

8. The working machine of claim 1, wherein at least part of the enclosure is mounted to the sidewall and/or wherein said at least part of the prime mover arrangement is mounted to the sidewall.

9. (canceled)

10. The working machine of claim 1, further comprising a duct within the enclosure configured to direct ambient air that passes through the inlet aperture towards said at least part of the prime mover arrangement.

11. The working machine of claim 1, wherein the enclosure comprises a first outlet aperture, wherein ambient air that passes through the inlet aperture, passes over said at least part of the prime mover arrangement, and then through the first outlet aperture; optionally, wherein the first outlet aperture comprises a grill for inhibiting debris ingress to the enclosure, optionally wherein the first outlet aperture extends through an external surface of the enclosure, wherein an angle between normals of the external surface and a ground plane of the working machine is in the range of 90 to 180 degrees.

12. (canceled)

13. The working machine of claim 11, wherein the first outlet aperture extends through an external surface of the enclosure, wherein an angle between a normal of the external surface and a forward travelling direction of the working machine is in the range of 90 to 180 degrees.

14. The working machine of claim 11, wherein the enclosure comprises a wheel housing for a ground engaging wheel of the ground engaging propulsion structure, and wherein the wheel housing comprises the first outlet aperture: optionally wherein the first outlet aperture faces the ground engaging wheel, and wherein the working machine comprises a barrier, such as a fender, interposed between the first outlet aperture and the ground engaging wheel for shielding the first outlet aperture from ground spray.

15. (canceled)

16. The working machine of claim 11, wherein ambient air that passes through the inlet aperture, over said at least part of the prime mover arrangement, and through the first outlet aperture, travels along a substantially linear path.

17. The working machine of claim 11, further comprising a second outlet aperture in the machine body and spaced from the enclosure, wherein ambient air that passes through the inlet aperture, then passes over at least part of the prime mover arrangement, and then through the second outlet aperture.

18. The working machine of claim 17, wherein the sidewall comprises an opening spaced from the inlet aperture, and wherein ambient air that passes through the inlet aperture, passes through said opening and then through the second outlet aperture, optionally wherein part of the prime mover arrangement (e.g. an electric motor) extends through said opening.

19. (canceled)

20. The working machine of claim 1, further comprising at least one blower within the enclosure configured to force ambient air towards said at least part of the prime mover arrangement, optionally wherein the at least one blower is adjacent the inlet aperture, and is selectively controllable so as to blow air out of the enclosure via the inlet aperture for blowing debris off a grill covering the inlet aperture.

21. (canceled)

22. The working machine of claim 1, wherein said at least part of the prime mover arrangement comprises one or more inverters housed within an inverter housing, and wherein ambient air that passes through the inlet aperture, passes through the inverter housing for cooling the one or more inverters.

23. The working machine of claim 1, wherein said at least part of the prime mover arrangement comprises at least one heatsink connected to at least one electrical component, such as an inverter, and wherein ambient air that passes through the inlet aperture, passes over said at least one heatsink; optionally, wherein the prime mover arrangement comprises at least two heatsinks, each connected to at least one electrical component of the prime mover arrangement, and wherein said at least two heatsinks are arranged in parallel with respect to a flow path of ambient air that passes through the inlet aperture, and over said at least part of the prime mover arrangement.

24. The working machine of claim 1, wherein ambient air that passes through the inlet aperture cools one or more of the following electrical components of the prime mover arrangement: one or more inverters; one or more electric motors; one or more electric pumps; and one or more power distribution units.

25. The working machine of claim 1, wherein the enclosure comprises: an access port for accessing the prime mover arrangement; and a hatch movable relative to the access port between open and closed positions for respectively opening and closing the access port, wherein the hatch inhibits precipitation ingress to the enclosure via the access port when in the closed position.
